
Recovery is not decision-rule validation: an equality audit and a failed abstention calibration

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Recovering a requested effect in simulation does not establish that an interval
2 or an abstention rule meets its operating requirements. We examine an instance
3 in which a single-cell recovery curve was used to justify a cell-count threshold.
4 Direct evaluation on two small colorectal single-cell cohorts does not support that
5 justification: the implemented pooled-cell simulation yields no qualifying cutpoint,
6 while reference-only draws produce candidates sensitive to count aggregation and
7 cohort composition. Reusing identical draws across grids isolates the contribution
8 of count support and binning. An equality audit explains the narrower question
9 answered by the original curve: its oracle estimate reproduces a realised empirical
10 reference. This equality does not cause the calibration failure or imply an invalid
11 estimator. A control gives the same zero-residual point estimate approximately
12 95% or 36% coverage by changing only its interval. Stratified patient and survival
13 simulations demonstrate how equivalent comparisons arise in trial-effect estimation
14 and depend on the reference, assignment design and observation process. Learned
15 generators preserve the identities in the tested saturated estimators; a limited search
16 of seven public repositories finds no functional reuse under its classification rule.
17 We propose reporting the reference and residual as an audit of what a comparison
18 measures, alongside separate evaluation of target error, interval performance and
19 abstention.

20 1 Motivation: recovery and decision-rule validation

21 A simulation can recover a requested treatment effect while leaving the clinical decision rule built
22 around that estimate untested. For virtual control arms, point accuracy, interval coverage and a
23 decision to withhold an estimate are distinct validation questions. Our motivating case is a single-cell
24 abstention rule justified by a recovery curve. We directly evaluate that rule, then use an equality
25 audit to identify what the earlier curve measured. The biological application supplies a documented
26 calibration failure; stratified patient simulations examine the corresponding comparison in trial-effect
27 estimation. Neither is a validation of a deployed virtual arm.

28 Our motivating application separates two explanations for differentiation-marker loss in colorectal
29 cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that
30 remain (an *intrinsic* change). These explanations suggest different biological interpretations. An
31 empirical comparison of per-cell means requires the relevant population to be observed in both tissues.
32 Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their
33 mean per-cell output. A reference-based decomposition, related to demographic standardisation and
34 Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

35 The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is
 36 undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells,
 37 the corresponding conditional mean itself has no referent. Substituting zero in either case turns an
 38 unavailable contrast into an apparent effect.

39 The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n <$
 40 50 , and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue.
 41 Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations
 42 examined below also retain raw arithmetic before this reporting gate. The question is whether
 43 the earlier recovery evidence established the interval-performance requirements used to justify the
 44 cutpoints.

45 **Relation to existing validation methods.** Simulation-study methodology already separates esti-
 46 mands, methods and performance measures [Burton et al., 2006, Morris et al., 2019]. Generator-
 47 analysis alignment is discussed in congeniality [Meng, 1994], plasmode and semi-synthetic bench-
 48 marks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019] and
 49 the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]; generators can also disadvantage
 50 good estimators [Shaw et al., 2026]. Our contribution is a case study of unsupported decision-rule
 51 validation, controlled analyses of its calibration sensitivity, and an operational audit of functional
 52 equivalence with trial-design examples. We do not propose a new estimator or infer invalidity from
 53 equality. Simulation-based calibration checks posterior ranks [Talts et al., 2018, Modrák et al., 2025];
 54 the point-estimator audit here neither replaces that check nor requires posterior samples. Appendix D
 55 provides further context.

56 2 Direct evaluation does not justify the abstention rule

57 **Target, interval and operating requirements.** The design document fixed a detectable halving of
 58 per-cell output ($s = 0.5$), interval inclusion of the stored reference in at least 90% of defined outputs,
 59 and discrimination of at least 80%, defined as the probability that the interval excludes zero at that
 60 effect. We retain the implementation’s term “coverage” for the first rate, with its target made explicit
 61 below. The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first
 62 meeting coverage alone. The committed values are 50 and 20. Candidates do not require every larger
 63 count to qualify and are not certified thresholds.

64 The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from
 65 GSE132465 (SMC) and six from GSE144735 (KUL3). We analyse GUCA2A in the differentiated
 66 population under fixed lineage labelling. Each draw samples, with replacement, 2,000 cells per arm
 67 from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool
 68 combines cells from both tissues before drawing either synthetic arm. Because the design document
 69 did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Neither reconstructs
 70 the real tissue contrast. A nominal 95% percentile interval resamples mature cells in both arms, with
 71 fractions held fixed and 200 inner draws unless stated otherwise. Undefined intervals are excluded
 72 from rate denominators and reported as abstentions (Appendix B).

73 The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

74 where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Thus the shift is fixed but this
 75 benchmark varies across draws; it is not the fixed source-pool parameter $f_N(s - 1)\mu_{\text{pool}}$. Coverage
 76 below means interval inclusion of Equation 2 under the existing calibration specification. Resampling
 77 both arms does not turn this into a conventional fixed-population coverage experiment. The results
 78 test the rule’s recorded justification; they do not establish nominal coverage for a patient population
 79 or provide a replacement clinical cutpoint.

80 **Controlled comparison of counts, bins and source pools.** Historical grids returned reference-
 81 only binned candidates of 100, 90 and 70, but changed random streams, bin boundaries and repli-
 82 cate budgets together (Appendix B). To isolate these choices, we simulated the union of counts
 83 once per cohort, pool and each of eight seeds, with 200 outer replicates and 200 inner bootstrap
 84 draws. Every grid summary reuses the same outputs at shared counts. The committed grid reaches
 85 $\{0, 20, 40, 100, 200, 400, 800\}$; filled and dense grids add count support. Fixed bins share boundaries

across grids; adaptive bins recalculate them. Binned candidates are median counts within the selected bin, not estimated crossing locations.

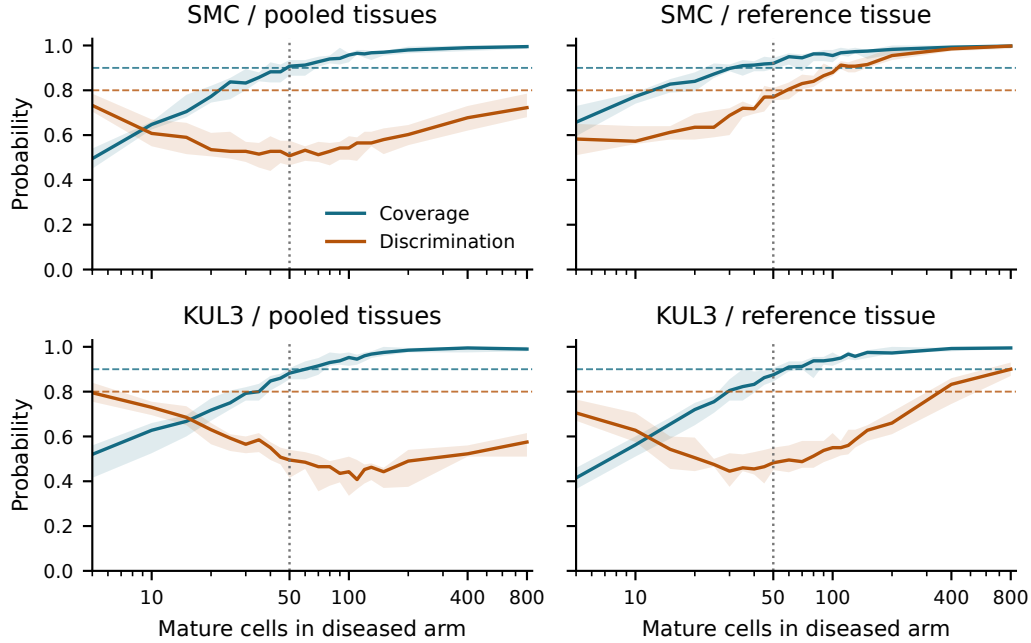


Figure 1: Direct evaluation on the controlled union of counts, without binning. Rows show cohorts; columns show draw pools. Curves are median rates over eight seeds and bands span their minima and maxima, conditional on the observed cohorts. Dashed lines mark the 90% coverage and 80% discrimination targets; the vertical line marks the committed count of 50. Rates condition on defined intervals and coverage uses the draw-dependent reference in Equation 2. Zero-cell settings have no defined interval and are omitted. No pooled-cell candidate qualifies. Per-count reference-only candidates vary by seed and cohort (Table 1).

With common fixed bins, SMC returns 100 on the committed grid and 70 on both expanded grids in all eight seeds. Adaptive bins return 65 on seven filled-grid seeds and 90 on one, while the dense grid returns 70 throughout. KUL3 instead returns 400 on six seeds and 800 on two, with all grid and binning schemes agreeing within each seed. Pooled draws yield no joint candidate in either cohort. The largest binomial Monte Carlo standard error is 0.035, conditional on the fixed cohorts; it does not measure patient-sampling uncertainty.

Table 1: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. These direct observations make the lack of support for a cohort-independent 50-cell rule more concrete than the selected binned candidates alone.

The per-count summaries in Table 1 supply a useful check on the bin medians. In SMC, the filled grid’s first qualifying actual count is 50 on two seeds, 60 on five and 80 on one; the dense grid adds a candidate of 45 on one seed. Thus a stable binned candidate of 70 is not evidence that 70 is a

uniquely determined biological threshold. A bin pools outcomes over several count settings, whereas a per-count rate addresses only that setting. Fixed bins hold the boundaries constant, but adding settings can still change both a bin’s median count and its aggregate performance.

These candidates select the first qualifying evaluated point, without fitting a monotone response or requiring all larger counts to qualify. Their adjacent crossing brackets describe grid support, not confidence intervals for a threshold. Conditional Monte Carlo errors quantify rate precision; they do not account for searching several counts and cannot certify a selected rule’s out-of-sample coverage. The controlled comparison establishes sensitivity of the recorded calibration procedure; it does not independently validate the selected rule.

Simulation precision is not patient-level uncertainty. Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort’s candidate at 70, but moves the second’s 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These are cohort-conditional influence checks, not intervals for a larger patient population. We therefore report candidates, not an endorsed calibrated cutpoint. Appendix B records further biological comparability limits.

Table 2: Reference-only sensitivity with every two-patient holdout pair represented once per count and effect. The three inner budgets reuse identical outer samples. Omission counts compare with the baseline at 200 inner draws; all pairs involving the omitted patient are removed and retained pairs are unchanged. These are sensitivity summaries within the observed cohorts.

cohort	pairs	joint candidate, inner draws			omissions changing candidate	
		200	1,000	5,000	joint	coverage only
SMC	45	70	70	70	0/10	5/10
KUL3	15	400	300	400	3/6	0/6

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Enumerating pairs improves transparency about source-patient influence, but overlapping pairs do not create additional independent patients.

The empirical finding is that the existing evidence does not justify a universal count of 50 under the specified calibration procedure. This result is separate from the equality audit below: neither zero nor nonzero residuals would determine whether a decision rule meets its operating requirements.

3 An equality audit of what recovery measures

The earlier internal analysis compared estimated intrinsic effects with the requested reference in Equation 2. Ratios were 1.00–1.07 where the rule returned an estimate on reference-only draws and 0.86–1.18 on pooled draws, with shrinking dispersion at larger counts (Appendix B). The oracle estimator applies Equation 1 to the draw’s own realised summaries. Its residual against that empirical decomposition is zero across 1,711,200 diagnostic replicates on two cohorts. The raw calculation also includes zero-cell cases; those arithmetic outputs do not define an intrinsic effect in an absent population (Appendix B). On defined contrasts, the recovery ratio therefore describes realised over requested reference. It does not measure whether the reporting gate meets its interval requirements.

The general form. Let D be a simulated draw requested at parameter θ . Pre-specify an observed-data reference $T(D) = h(S(D))$, where S denotes summaries of the draw. We call $T(D)$ the

142 *realised reference*; it need not be a realised causal effect. If the estimate uses the same functional,
 143 $\hat{\theta}(D) = h(S(D))$, then, for $\theta \neq 0$,

$$\frac{\hat{\theta}(D)}{\theta} = \frac{T(D)}{\theta}, \quad r(D) = \hat{\theta}(D) - T(D) \equiv 0.$$

144 `r = estimate[valid] - reference[valid]`
 145 `max_residual = np.max(np.abs(r))`

146 Here `valid` marks paired finite outputs; report excluded draws and do not compute the maximum if
 147 none remain. A maximum below an explicitly chosen tolerance in outcome units flags numerical
 148 equality for investigation. Only an algebraic argument establishes an identity. Sharing summaries
 149 alone is insufficient: different functions of the same summaries need not agree.

150 **What the audit can establish.** The recovery curve still measures sampling performance against θ . If
 151 estimate and reference are equivalent, their curves are redundant for comparing those implementations.
 152 That is compatible with an accurate estimator and a useful recovery experiment. The unsupported
 153 inference is from this point-estimate comparison to interval coverage or an abstention rule, neither
 154 of which the comparison evaluates. A zero residual therefore prompts inspection of the claimed
 155 evidence; it is not a rejection criterion for an estimator.

156 **A control separating equality from performance.** To test this distinction directly, a separate
 157 simulation fixes the reference before evaluating any estimator: for 200 independent observations
 158 $Y_i \sim \mathcal{N}(\theta, 1)$, set $T(D) = \bar{Y}$. We evaluate four procedures at $\theta = 0$ and 0.5, with 2,000 replicates
 159 per effect. The baseline reports the full-sample mean with a normal 95% interval using the sample
 160 standard deviation. A second procedure keeps that point estimate but quarters the interval half-width.
 161 A third adds a known bias of 0.5 to the baseline point and interval. The fourth uses every other
 162 observation, selected before seeing outcomes, with its own mean and standard error. This half-sample
 163 estimate shares observations with the full-sample reference; it is not independent of $T(D)$.

Table 3: Equality and statistical performance in the clean control. All 2,000 outputs per row are finite. Residuals compare points with the pre-specified full-sample mean; bias, RMSE and coverage compare with the requested θ . The nominal coverage is 95%.

θ	procedure	$\max r $	bias	RMSE	coverage
0	Full-sample mean	0	-0.0008	0.0710	94.60%
	Same point, narrow interval	0	-0.0008	0.0710	35.95%
	Mean plus 0.5	0.5000	0.4992	0.5042	0.00%
	Half-sample mean	0.2883	0.0004	0.0994	95.00%
0.5	Full-sample mean	0	0.0001	0.0710	95.40%
	Same point, narrow interval	0	0.0001	0.0710	36.30%
	Mean plus 0.5	0.5000	0.5001	0.5052	0.00%
	Half-sample mean	0.2686	-0.0017	0.1015	94.20%

164 Table 3 puts adequate and inadequate intervals in the same zero-residual class. Conversely, departure
 165 contains both the unbiased half-sample mean and the deliberately biased estimate. Coverage Monte
 166 Carlo standard errors are at most 1.08 percentage points. Neither residual class certifies performance,
 167 even with a reference declared in advance. At the null, differences remain well-defined although
 168 recovery ratios do not.

169 **Report performance and equivalence separately.** State the scientific target and whether it is fixed
 170 or draw-dependent, then report target error and interval performance. If a realised reference is also
 171 used, declare its functional and observation process before inspecting residuals. Report attempted and
 172 valid pairs, maximum absolute residual and an explicit absolute tolerance in outcome units. Establish
 173 identities algebraically on a stated class of datasets; agreement on a restricted, noiseless design need
 174 not extend beyond it. Evaluate abstention frequency and conditional interval performance separately,
 175 using independent evaluation draws for any selected rule. At null effects, report differences instead
 176 of undefined recovery ratios.

177 4 Reference dependence in trial-effect simulations

178 The case study concerns a biological decision rule; the following experiments examine the same
 179 distinction in stylised trial-effect estimation. Their targets are fixed generating parameters. These
 180 simulations establish possible identities, not their frequency in clinical practice.

181 **A stratified patient simulator.** For stratum counts n_g , total sample size n , and observed arm means
 182 $\bar{Y}_{ag}$, define

$$T(D) = \sum_g \frac{n_g}{n} (\bar{Y}_{1g} - \bar{Y}_{0g}).$$

183 Saturated G-computation [Robins, 1986] calculates this contrast directly. IPW with empirical
 184 propensities $\hat{p}_g = n_{1g}/n_g$ also equals it: each stratum's treated weighted sum divided by n is
 185 $(n_g/n)\bar{Y}_{1g}$, and likewise for controls. This requires nonempty arms within each stratum and these
 186 specific weights. It is not an equivalence for arbitrary outcome or propensity models. The reference
 187 is an observed contrast, not the finite-sample mean of paired potential-outcome differences.

188 Figure 2 evaluates five estimators in three covariate strata with stratum-dependent treatment probabili-
 189 ties, a homogeneous additive effect $\theta = 3$, and cohorts of 100–5,000 patients. Four estimators recover
 190 the requested effect and tighten with cohort size. Two reproduce $T(D)$: G-computation has exactly
 191 zero residual and saturated IPW agrees to numerical precision. Cross-fitting the IPW propensity
 192 [Chernozhukov et al., 2018] breaks the equality in this design because each patient's weight is fitted
 193 on a different fold. It remains consistent here, but its nonzero residual establishes only departure from
 194 this reference. The unadjusted contrast is confounded, with a recovery ratio near 2.4; the recovery
 195 curve correctly reveals that error.

196 Ordinary least squares (OLS) with treatment and additive stratum dummies also departs from $T(D)$,
 197 although it uses every record and is correctly specified for this homogeneous effect. Standardisation
 198 weights strata by prevalence; OLS uses normalised weights proportional to $n_g\hat{p}_g(1 - \hat{p}_g)$. Its residual
 199 is smaller than cross-fitted IPW's at every tested cohort size. Thus the audit concerns a shared
 200 functional, not simply a shared sample.

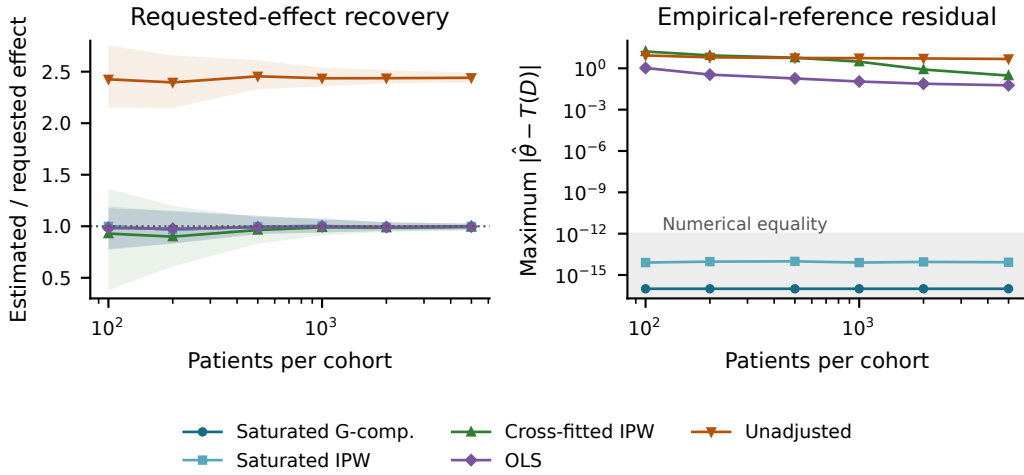


Figure 2: Patient simulation with confounded assignment and requested effect $\theta = 3$. Left: median and IQR of recovery over 200 replicates (197 at the smallest cohort, after excluding three draws with an empty stratum-arm cell). Right: residual against the standardised observed-data reference on a log scale, with a band at machine precision. G-computation and saturated IPW reproduce the reference. OLS and cross-fitted IPW depart from it while recovering the requested effect. The unadjusted contrast is confounded. Departure is not a validity criterion.

201 **Assignment and the chosen reference change the equality.** The sweep crosses 14 estimators,
 202 three designs, two realised references, six cohort sizes and six seeds (settings in Appendix A). Table 4
 203 shows nine of those estimators under two of the designs. Independent Bernoulli assignment is

discussed below; the five logistic-propensity variants are reported in Appendix C. Under confounded assignment, OLS has maximum residual 0.0663 against the standardised reference at $n = 5,000$, but 3.5×10^{-14} against the variance-weighted reference equal to OLS by the Frisch–Waugh identity. G-computation reverses the comparison: 0 and 0.0663, respectively. Both consistently estimate the requested homogeneous effect.

With complete, stratified permuted blocks and 1:1 allocation, empirical propensities are 0.5 in every stratum. The two references coincide, and OLS and even the unadjusted contrast reproduce them to numerical precision (1.6×10^{-14} and 6.7×10^{-15}). Independent Bernoulli assignment at probability 0.5 does not enforce this realised balance: at $n = 5,000$, OLS and the unadjusted contrast have maximum residuals 9.51×10^{-4} and 0.562, respectively, against the standardised reference. Thus 1:1 randomisation in expectation is insufficient for the identity. Non-cross-fitted AIPW with saturated nuisances also equals standardisation: its within-stratum augmentation sums vanish (residual 4.4×10^{-15} in the confounded design). Exact matching using all controls within each stratum agrees to 8.9×10^{-16} ; using 20 or one control gives residuals of 0.052 and 0.164. These are properties of the implemented estimator, reference and design together. Appendix C also examines regularisation and effect heterogeneity.

Table 4: Maximum absolute residual at 5,000 patients, with 1,200 replicates pooled over 6 seeds. Values near 10^{-14} reflect numerical equality. Changing the reference or using complete 1:1 blocks changes which estimators reproduce the reference. These maxima use more replicates than Figure 2, so their nonzero magnitudes are not directly comparable.

estimator	confounded		block-randomised	
	standardised	var-weighted	standardised	var-weighted
G-computation, generator’s strata	0	0.0663	0	9×10^{-16}
IPW, saturated propensity	1×10^{-14}	0.0663	7×10^{-15}	7×10^{-15}
AIPW, saturated, no cross-fitting	4×10^{-15}	0.0663	4×10^{-15}	4×10^{-15}
matching on stratum, all controls	9×10^{-16}	0.0663	9×10^{-16}	0
matching on stratum, 20 controls	0.052	0.067	0.037	0.037
matching on stratum, 1 control	0.164	0.182	0.170	0.170
OLS, stratum dummies	0.0663	3.5×10^{-14}	2×10^{-14}	2×10^{-14}
IPW, cross-fitted	0.599	0.618	0.182	0.182
unadjusted difference	4.99	4.99	7×10^{-15}	7×10^{-15}

Censoring changes the comparison. We repeated the patient experiment with exponential event times, confounded assignment and administrative censoring: four censoring levels, seven cohort sizes, and 1,200 replicates pooled over 6 seeds per cell. Paired comparisons exclude non-finite outputs, including event-free cells and Cox fitting failures. Exclusions occur even at low censoring and remove 463 of 1,200 draws at 78% censoring and $n = 100$; reported residuals are conditional on valid pairs.

The per-stratum exponential maximum-likelihood estimator uses rates equal to events divided by exposure, then averages log rate contrasts by stratum prevalence. Its observed-data reference is that same functional, giving maximum residual zero in all 28 cells. Against the corresponding latent-time reference, maximum residuals at $n = 3,000$ are 0, 0.061, 0.144 and 0.376 at 0%, 12%, 49% and 78% censoring. The worst values across cohort sizes are 0, 0.473, 1.089 and 1.586. The estimator is unchanged; the latent comparison measures a different departure, associated with information hidden by censoring.

At 49% censoring, the exponential estimator’s maximum latent-reference residual exceeds stratified Cox’s in five of seven cohort sizes; at 78%, in six of seven (0.376 versus 0.293 at $n = 3,000$). Per-replicate inversion rates are 0.53–0.61. Neither ordering is a quality criterion: a larger residual cannot be read as a more informative validation. Use an observed-data reference for the reuse audit, and retain requested parameters and latent targets for their separate error, coverage and information-loss evaluations.

A restricted-mean survival time (RMST) comparison provides a control. Its horizon is fixed at $\tau = 2$, below every administrative censoring time in the sweep. Standardised Kaplan–Meier RMST therefore reproduces the empirical latent truncated-mean contrast to numerical precision across all 28 cells (maximum residual 2.8×10^{-15} on valid outputs). Censoring has hidden no information within this

horizon. The observation process matters through its effect on the chosen functional, not simply through the fraction censored.

An independent check of eight Cox fits agrees with `lifelines` to 1.4×10^{-7} , below the specified 10^{-5} tolerance (Appendix C). It checks implementation agreement on one cohort, not coverage or convergence across the sweep.

Learned generators preserve the tested identities. A further sweep fits Gaussian-mixture and MLP generators, samples synthetic cohorts, and refits estimators on each cohort. Saturated G-computation exactly reproduces the drawn reference in all 30 learned-generator and 12 parametric control settings. This is an estimator identity that holds regardless of how the cohort was generated; it does not establish generator fidelity. Penalised outcome models instead yield nonzero residuals. Of 420 estimator-by-generator-by-size settings, 41 pass a relative-plus-absolute numerical comparison but fail an absolute-only one; all 41 use ridge penalties. These are tolerance classifications, not evidence of worse statistical performance with more data. Appendix C reports the full design and the distinct model-based reference; Appendix C discusses residual scale.

Table 5: Maximum absolute residual against the drawn reference, over cohort sizes and generators within each family, conditional on finite pairs. All ten estimators in the expanded run are shown. The saturated zero row is exact; other entries are rounded from the pinned primary table. Changing generator capacity preserves the saturated identities but can change other residuals.

Estimator	Mixtures	MLP generator	Parametric controls
G-computation, saturated	0	0	0
IPW, saturated	1.6×10^{-14}	1.2×10^{-14}	1.1×10^{-14}
G-comp., mixture outcome $K = 1$	9.3×10^{-15}	8.0×10^{-15}	8.0×10^{-15}
G-comp., mixture outcome $K = 4$	2.2×10^{-14}	1.6×10^{-14}	1.8×10^{-14}
G-comp., ridge $\alpha = 10^{-5}$	7.9×10^{-5}	6.3×10^{-5}	8.6×10^{-5}
G-comp., ridge $\alpha = 10^{-3}$	0.0079	0.00627	0.00864
G-comp., ridge $\alpha = 10^{-1}$	0.713	0.568	0.78
G-comp., MLP outcome	3.3	0.00908	0.0125
OLS, stratum dummies	1.51	1.42	1.71
Unadjusted difference	10.2	9.69	11.3

Table 5 reports every estimator in the expanded run, including the MLP outcome model whose maximum residual reaches 3.3 on mixture-generated data. Saturated and mixture-outcome identities therefore should not be extrapolated to arbitrary learned estimators. In a mixture M-step, $\pi_k = n_k/n$ and $\mu_k = \sum_i r_{ik} y_i / n_k$ imply $\sum_k \pi_k \mu_k = \bar{y}$; generator complexity cannot break this particular mean-preservation identity. By contrast, numerical closeness depends on the chosen tolerance: comparing residuals to zero uses absolute tolerance alone, whereas a relative tolerance additionally scales with the reference magnitude. The recorded run contains 75,580 estimator evaluations and 74,726 finite drawn-reference pairs; Appendix C reconciles the exclusions.

Evidence outside the constructed examples is limited. A pre-registered search classified seven public repositories and found no functional reuse under its rule. Six compare with generating parameters or quantities computed outside the observed sample; the remaining case involves distinct functionals that agree on noiseless linear data. Adding outcome noise separates them, illustrating the need to state the domain of an identity. The search did not locate examinable code for its query specifically targeting virtual control arms. This is neither a representative prevalence estimate nor evidence that deployed systems are affected (Appendix D).

5 Conclusion

Direct evaluation does not support the original cell-count rule’s recorded justification. The pooled simulation yields no qualifying candidate; reference-only candidates depend on cohort and aggregation, with additional sensitivity to patient omission and bootstrap budget. The evaluation uses a draw-dependent reference and does not certify fixed-population coverage or a replacement clinical threshold.

The equality audit answers a separate question: whether the estimator reproduces a specified empirical functional. Our simulations show that this can hold for valid estimators and depends on the reference, trial design and observation process. It neither causes the calibration failure nor establishes a flaw in an estimator. Report target error, interval performance and abstention directly; use functional-equivalence checks to make the scope of recovery evidence explicit. No new patient outcome model or biological mechanism is validated by these results.

6 Responsible use, limitations and impact

Scope and interpretation. This paper makes methods and validation claims, not biological or clinical claims. The pre-committed control panel included genes expected to change compositionally, genes expected to change intrinsically, and a retained-gene control. In the primary cohort, the retained-gene control changed by -1.000 to -0.951 relative to baseline, almost matching the compositional controls' -1.000 to -0.952 range. The intrinsic control was too weakly expressed to provide a useful test (0.04 CP10K versus 19.4 and 24.9 for the two abundant compositional controls; CP10K denotes counts per 10,000). The controls therefore did not establish the intended mechanistic separation. A second cohort reproduced the pattern, without resolving its interpretation.

Misuse and limitations. NOT ESTIMABLE must not be read as “no effect” or as evidence that surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B). Calibration here covers one gene, one labelling axis and one weighting on two small cohorts. It challenges this rule’s justification, not cell-count rules in general. Its stored coverage reference depends on the sampled baseline, as specified in Equation 2. The draw-pool choice was underspecified, repeated simulations do not add independent patients, and the leave-one-patient-out sensitivity remains conditional on these cohorts; patient-level population uncertainty in the candidate cutpoints is unquantified. The absence-defined labelling axis remains sensitive to sequencing depth (Appendix B). The patient simulations illustrate possible functional equalities; they do not estimate their prevalence in deployed virtual control arms.

Data and impact. The single-cell cohorts are public, de-identified and previously published. The release policy covers derived summaries and code, rather than cell-level matrices. A system reporting group-specific quantities should distinguish an unavailable comparison from an estimated null and should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration attempts helps prevent unsupported thresholds from being treated as validated.

References

- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.
- Alan S. Blinder. Wage discrimination: Reduced form and structural estimates. *The Journal of Human Resources*, 8(4):436–455, 1973.
- Andrea Burton, Douglas G. Altman, Patrick Royston, and Roger L. Holder. The design of simulation studies in medical statistics. *Statistics in Medicine*, 25(24):4279–4292, 2006. doi: 10.1002/sim.2673.
- Maren Büttner, Johannes Ostner, Christian L. Müller, Fabian J. Theis, and Benjamin Schubert. sccoda is a bayesian model for compositional single-cell data analysis. *Nature Communications*, 12(1):6876, 2021.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.
- Helena L. Crowell, Charlotte Soneson, Pierre-Luc Germain, Daniela Calini, Ludovic Collin, Catarina Raposo, Dheeraj Malhotra, and Mark D. Robinson. muscat detects subpopulation-specific state transitions from multi-sample multi-condition single-cell transcriptomics data. *Nature Communications*, 11(1):6077, 2020.
- Alicia Curth, David Svensson, James Weatherall, and Mihaela van der Schaar. Really doing great at estimating CATE? A critical look at ML benchmarking practices in treatment effect estimation. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.
- Emma Dann, Neil C. Henderson, Sarah A. Teichmann, Michael D. Morgan, and John C. Mariotti. Differential abundance testing on single-cell data using k-nearest neighbor graphs. *Nature Biotechnology*, 40(2):245–253, 2022.
- Ran El-Yaniv and Yair Wiener. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11:1605–1641, 2010.
- Jessica M. Franklin, Sebastian Schneeweiss, Jennifer M. Polinski, and Jeremy A. Rassen. Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases. *Computational Statistics & Data Analysis*, 72:219–226, 2014.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.
- Amanda Gentzel, Dan Garant, and David Jensen. The case for evaluating causal models using interventional measures and empirical data. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 11717–11727, 2019.
- Jari Kaipio and Erkki Somersalo. Statistical inverse problems: discretization, model reduction and inverse crimes. *Journal of Computational and Applied Mathematics*, 198(2):493–504, 2007.
- Evelyn M. Kitagawa. Components of a difference between two rates. *Journal of the American Statistical Association*, 50(272):1168–1194, 1955.
- Hae-Ock Lee, Yourae Hong, Hakki Emre Etlioglu, Yong Beom Cho, et al. Lineage-dependent gene expression programs influence the immune landscape of colorectal cancer. *Nature Genetics*, 52:594–603, 2020. doi: 10.1038/s41588-020-0636-z.
- Xiao-Li Meng. Multiple-imputation inferences with uncongenial sources of input. *Statistical Science*, 9(4):538–558, 1994.

354 Martin Modrák, Angie H. Moon, Shinyoung Kim, Paul-Christian Bürkner, Niko Huurre, Kateřina
355 Faltejsková, Andrew Gelman, and Aki Vehtari. Simulation-based calibration checking for Bayesian
356 computation: the choice of test quantities shapes sensitivity. *Bayesian Analysis*, 20(2):461–488,
357 2025.

358 Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical
359 methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi: 10.1002/sim.8086.

360 Ronald Oaxaca. Male-female wage differentials in urban labor markets. *International Economic*
361 *Review*, 14(3):693–709, 1973.

362 Viktor Petukhov, Anna Igolkina, Rasmus Rydbirk, Shenglin Mei, Lars Christoffersen, Konstantin
363 Khodosevich, and Peter V. Kharchenko. Case-control analysis of single-cell rna-seq studies.
364 *bioRxiv*, page 2022.03.15.484475, 2022. doi: 10.1101/2022.03.15.484475.

365 Belinda Phipson, Choon Boon Sim, Enzo R. Porrello, Alex W. Hewitt, Joseph Powell, and Alicia Osh-
366 lack. propeller: Testing for differences in cell type proportions in single cell data. *Bioinformatics*,
367 38(20):4720–4726, 2022.

368 James Robins. A new approach to causal inference in mortality studies with a sustained exposure
369 period—application to control of the healthy worker survivor effect. *Mathematical Modelling*, 7
370 (9-12):1393–1512, 1986.

371 Pamela A. Shaw, Susan Gruber, Brian D. Williamson, Rishi J. Desai, Susan M. Shortreed, Chelsea
372 Krakauer, Jennifer C. Nelson, and Mark J. van der Laan. A cautionary note for plasmid simulation
373 studies in the setting of causal inference. *Statistics in Medicine*, 2026.

374 Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating
375 Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*,
376 2018.

377 Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random*
378 *World*. Springer, 2005.

379 Armand Wirgin. The inverse crime. *arXiv preprint math-ph/0401050*, 2004.

380 Jun Zhao, Ariel Jaffe, Henry Li, Ofir Lindenbaum, Esen Sefik, Ruaidhrí Jackson, Xiuyuan Cheng,
381 Richard A. Flavell, and Yuval Kluger. Detection of differentially abundant cell subpopulations in
382 scrna-seq data. *Proceedings of the National Academy of Sciences*, 118(22):e2100293118, 2021.

383 A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. Coverage uses $f_N(s-1)\bar{Y}_N$ with the drawn reference-arm mean, not a fixed pool mean.
Patient outcomes	Stratum probabilities (0.5, 0.3, 0.2), baseline means (10, 14, 25) and $Y \mid (g, A) \sim \mathcal{N}(\mu_g + 3A, 4^2)$. Confounded treatment probabilities (0.25, 0.50, 0.75); alternatives are independent Bernoulli assignment at 0.5 and complete balanced blocks of size 4 within strata. The five-method curve uses 200 replicates per size; the three-design sweep uses 200 per seed across six seeds.
Survival outcomes	The same stratum and confounded-assignment probabilities; exponential event rate $\lambda_g \exp(\beta A)$, $\lambda = (0.10, 0.16, 0.30)$, $\exp(\beta) = 0.6$. Administrative horizons 1000, 20.065216, 6.072920, 2.028858 target 0%, 12%, 49%, 78% censoring. The same latent cohort is viewed through each horizon; RMST uses $\tau = 2$. Each size has 200 replicates per seed over six seeds.

385 Settings are recorded by `calibration_gap`, `trial_recovery` and `trial_survival`; the dense
386 calibration grid is specified by `cutpoint_dense_grid`. Historical grid budgets are reported in
387 Appendix B; Section 2 specifies the controlled runs. These are existing simulations, not clinical-trial
388 data.

389 B Calibration details and limits

390 **Historical grids.** Before the controlled run, the SMC reference-only sweeps returned binned
391 joint candidates of 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and
392 200 replicates per setting, respectively. Their adjacent crossing brackets were (40, 100], (65, 90]
393 and (42.5, 70], describing evaluated bin summaries rather than confidence intervals. The 100-to-90
394 change occurred within the same bin, (76, 107]: adding counts 80 and 90 to 100 changed both its
395 median count and its aggregated outcomes. Random streams and bin boundaries also differed, so this
396 comparison alone did not isolate a resolution effect. The controlled analysis in Section 2 supersedes
397 it for that question. Under pooled draws no joint cutpoint qualified in either cohort’s historical grids.
398 KUL3 reference-only candidates were 400 or 800; its largest setting sampled 800 mature cells per
399 arm with replacement from 844 distinct mature source cells.

400 **Recovery accounting and unavailable contrasts.** The four diagnostic sweeps comprise 1,711,200
401 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested
402 reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned
403 into 89,700 and 29,900, respectively. The maximum estimate–realised-reference residual is zero
404 throughout. At a zero-cell setting the raw ratio is $1/(1-s)$ when the denominator is nonzero. These
405 raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an
406 effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined
407 intervals were previously counted as failures through comparisons with NaN; corrected rates exclude
408 them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed
409 gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted
410 regions.

411 **Biological scope and patient uncertainty.** The finest annotation resolution yields abstention in
412 28/28 patients under one population definition and 24/28 under the other; coarser definitions do
413 not resolve identifiability. The coarsest definition labels every epithelial cell mature and forces the
414 compositional term to zero. In a separate comparability analysis over three cohorts and 42 patients,
415 83 of 84 housekeeping-detection control rows pass a 0.10 tolerance despite expression changes of

1.19–1.64 fold. Patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort, leaving the comparisons undecided. Depth matching discards roughly three quarters of the cells and pushes about half the remaining estimable rows below the count threshold. The two source pools also differ jointly in tissue mixture, dispersion and distinct mature-cell counts (662 versus 3,824); their comparison cannot isolate these factors. These observations restrict biological interpretation. Repeated cell draws and overlapping patient pairs do not supply independent patient replication.

C Supporting trial and generator checks

Reference dependence beyond the main comparison. For logistic-propensity IPW, confounded-design maximum residuals against the standardised reference are 0.403, 0.045, 0.0103 and 0.010 at $C = 0.1, 1, 100, 10^6$; tightening solver tolerance to 10^{-8} reduces the last to 1.3×10^{-6} . Under increasing effect heterogeneity ($\tau = 0$ to 16, fixed population ATE), OLS recovery grows from 1.003 to 1.132, while its residual against its own variance-weighted reference stays at numerical precision. Recovery can therefore reveal target mismatch even when functional equality persists. These results supplement Table 4.

Residual scale. The optional descriptor $\rho = \text{med}|\hat{\theta} - T| / \text{med}|T - \theta|$ compares typical departure with reference error when its denominator is positive. It is zero at exact equality and undefined when the denominator is zero. Because the two terms in $\hat{\theta} - \theta = (\hat{\theta} - T) + (T - \theta)$ can covary or cancel, ρ neither decomposes total error nor ranks estimators. A zero median can also conceal nonzero residuals, so it cannot replace the maximum-residual check.

Independent Cox implementation check. Eight stratified and unadjusted Cox fits on one fixed 1,500-patient cohort at four censoring horizons agree with `lifelines.CoxPHFitter` to 1.4×10^{-7} , below the specified 10^{-5} tolerance. All eight pairs are finite. This checks the implemented coefficients on that cohort, not coverage or convergence over the simulation grid.

Learned-generator design and exclusions. The pre-registration preceded the module; a recorded amendment added penalised outcome models after the original estimator set could not test the proposed near-equality interval. Seven generators (mixtures at $K = 1, 2, 4, 8$, an MLP conditional Gaussian and two parametric controls), ten estimators, six cohort sizes, six seed streams and 30 replicates schedule 75,600 evaluations on 7,560 synthetic cohorts (seed 20260913). After generator failures the saved table contains 75,580 evaluations on 7,558 cohorts. Against the drawn reference, 74,726 pairs are finite and 854 are excluded: 410 have nonfinite references and 731 nonfinite estimates, overlapping in 287 evaluations. All 420 generator-by-estimator-by-size cells retain valid pairs. Table 5 uses maxima within each generator family, conditional on those pairs.

Tolerance and the model-based target. NumPy’s default comparisons use 10^{-8} against a zero residual and $10^{-8} + 10^{-5}|T|$ for estimate versus reference. Thirteen generator–estimator combinations fail both at $n = 100$ but pass only the second at $n = 5,000$; all use ridge penalties. In the saturated ridge model, fitted means are $\bar{y}_{ga}n_{ga}/(n_{ga} + \alpha)$, so a fixed penalty diminishes with cell count. This numerical transition does not imply declining statistical validity. The fitted generator’s population effect T_{model} is a different reference from the empirical $T(\tilde{D})$. Mixture references use 200,000 Monte Carlo observations; saved median standard errors are 0.019–0.027. Parametric and MLP references are analytic. In $\hat{\theta} - \theta = (\hat{\theta} - T_{\text{model}}) + (T_{\text{model}} - \theta)$, estimator departure and generator target discrepancy can covary. The finite Monte Carlo budget limits numerical resolution, not the target’s intrinsic precision; neither reference validates a model for real patients.

D External audit and research context

Search and classification. A pre-registered queue contained 41 entries; 19 were screened out and seven classified within the 26 entries examined in order. An amendment before code inspection relaxed unsuccessful queries; the classification rule was unchanged. Generating-parameter references or quantities computed outside the estimator’s observed records give NO. Same-record references require examination of inputs, weights, conditioning, nuisance fitting and observation process; YES

465 requires affirmative evidence of functional identity. All seven recorded verdicts are NO: `causal`, `target-`
466 `trial-emulation`, `ecasim`, `PlasmodeSimulation`, `StatsPAI`, `diff-diff`, and `Auto-Empirical-Research-`
467 `Skills`. The first six use generating parameters or outside-sample quantities; the last compares
468 cross-fitted partialling-out with full-sample OLS. Its 4.0×10^{-14} residual on noiseless linear data
469 becomes 0.125 after adding outcome noise. The functions agree on the restricted noiseless support but
470 not on general noisy datasets; the NO verdict concerns shared functional form. Classification used one
471 reader, only this last repository was executed, and the sample is nonrandom. The virtual-control-arm
472 query located no examinable code. These limits preclude a prevalence estimate.

473 **Related scope.** Cell-count filters are established in pseudobulk analysis [Crowell et al., 2020];
474 `scCODA`, `Milo`, `propeller` and `DA-seq` address composition [Büttner et al., 2021, Dann et al., 2022,
475 Phipson et al., 2022, Zhao et al., 2021], while `cocoa` separates composition and expression changes
476 [Petukhov et al., 2022]. Selective prediction motivates evaluating both answers and abstentions
477 [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Conformal prediction
478 coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] does not define a missing conditional
479 population or automatically provide intervals for this intrinsic parameter.

480 **Implementation and provenance.** Corrections separated population definitions collapsed by de-
481 duplication, derived fractions and means from one labelling call, tracked producing scripts, and
482 repaired undefined-interval denominators. Saved tables retain commit hashes and seeds. These
483 implementation checks support traceability, not statistical validity; the draw-dependent calibration
484 target is stated in Equation 2.

485 **Use of large language models.** Language-model assistants supported coding, figures, drafting and
486 model audits under author direction. They helped identify defects and introduced some of those
487 above. Authors remain responsible for all claims. Automated checks re-derive selected results and
488 complement review; they do not verify every manuscript statement.